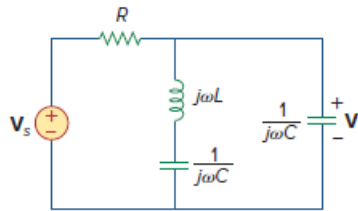


Problem

Using nodal analysis obtain $\mathbf{V}$ in the circuit of Fig. 10.72.

Figure 10.72:



Step-by-step solution

Step 1 of 1

$$\frac{V - V_s}{R} + \frac{V}{\left(j\omega L + \frac{1}{j\omega C}\right)} + j\omega C V = 0$$

$$V + \frac{j\omega R C V}{-\omega^2 L C + 1} + j\omega R C V = V_s$$

$$\left[\frac{1 - \omega^2 L C + j\omega R C + j\omega R C - j\omega^3 R L C^2}{1 - \omega^2 L C} \right] V = V_s$$

$$V = \left[\frac{(1 - \omega^2 L C) V_s}{1 - \omega^2 L C + j\omega R C (2 - \omega^2 L C)} \right]$$